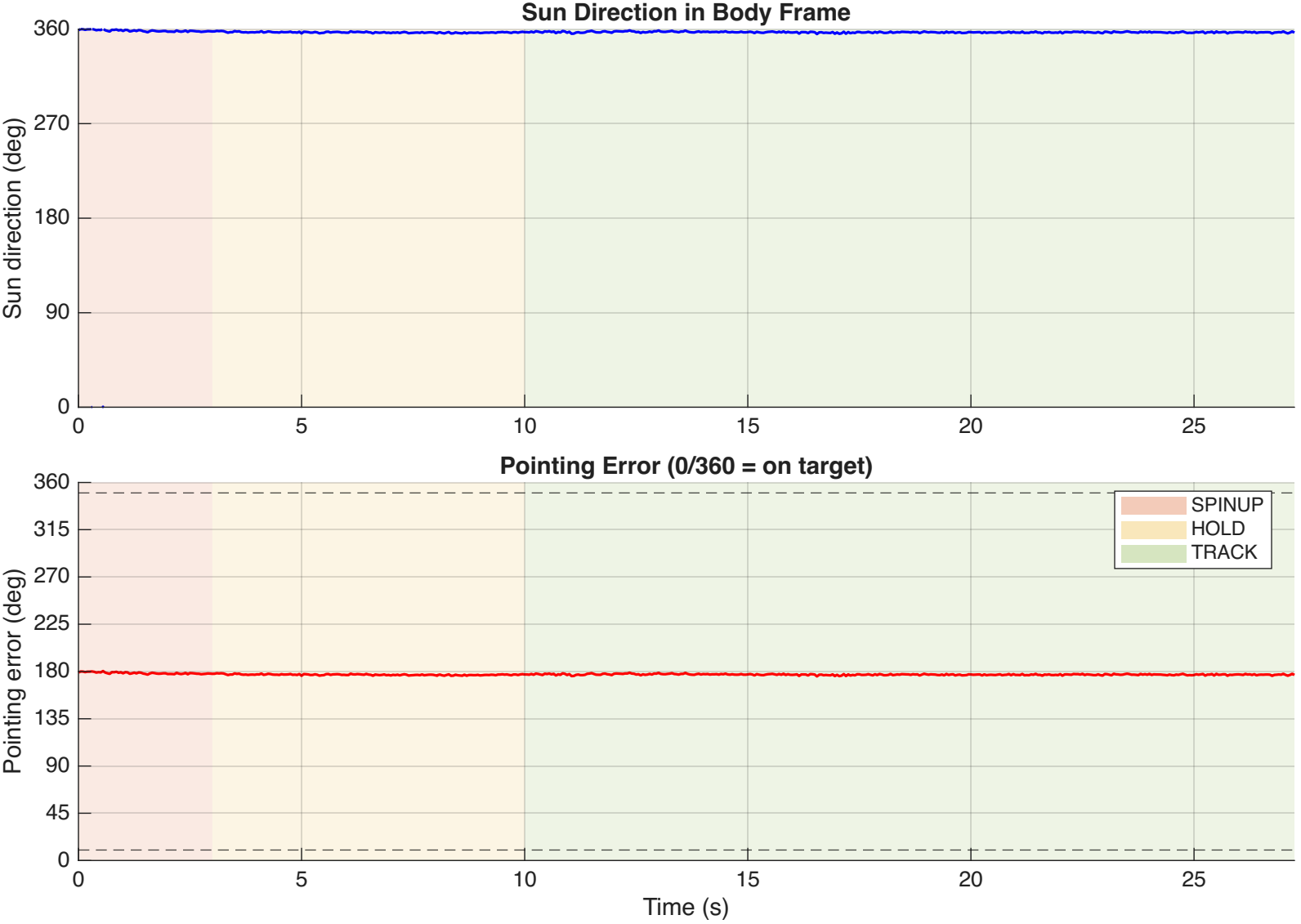
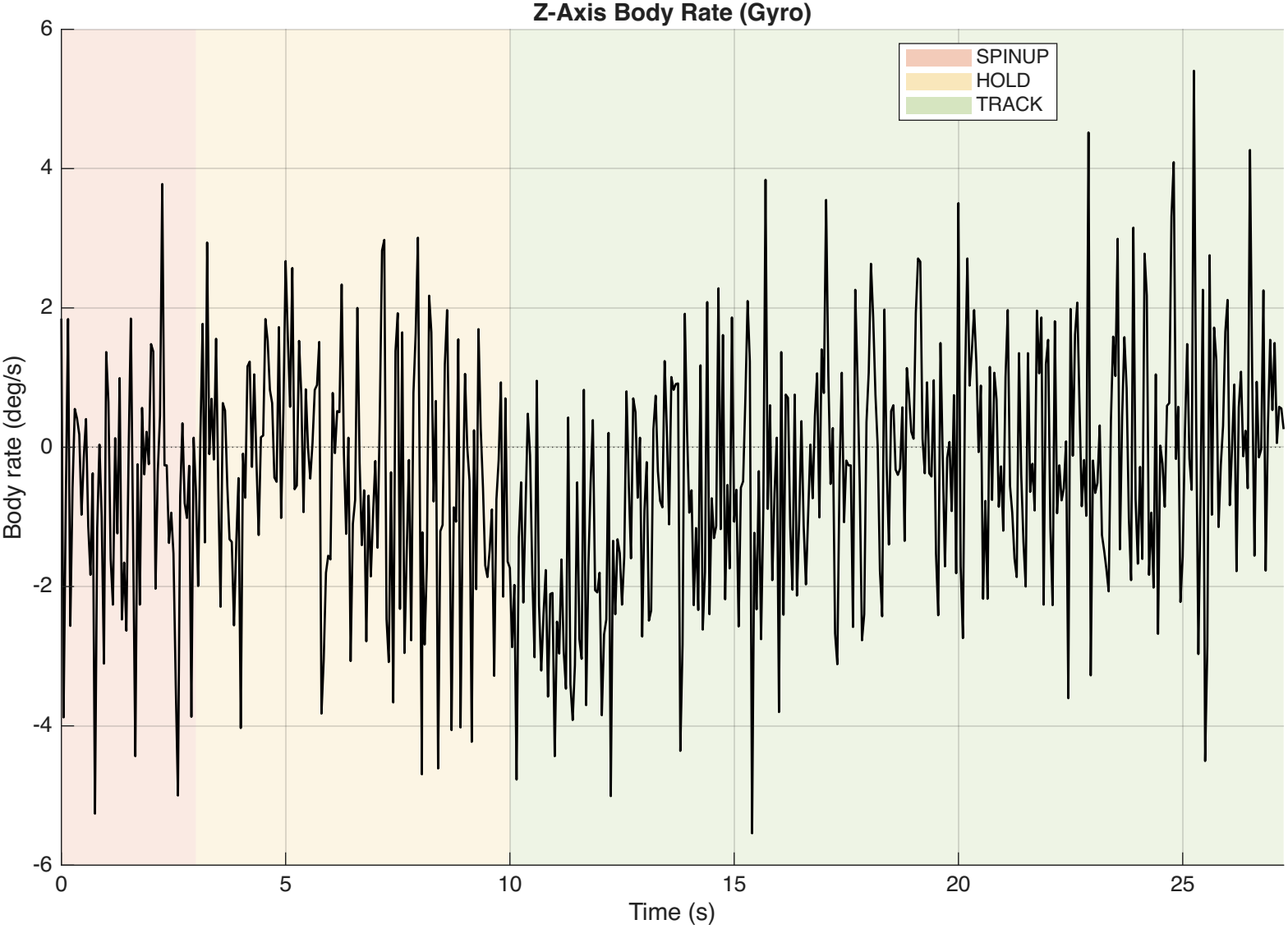
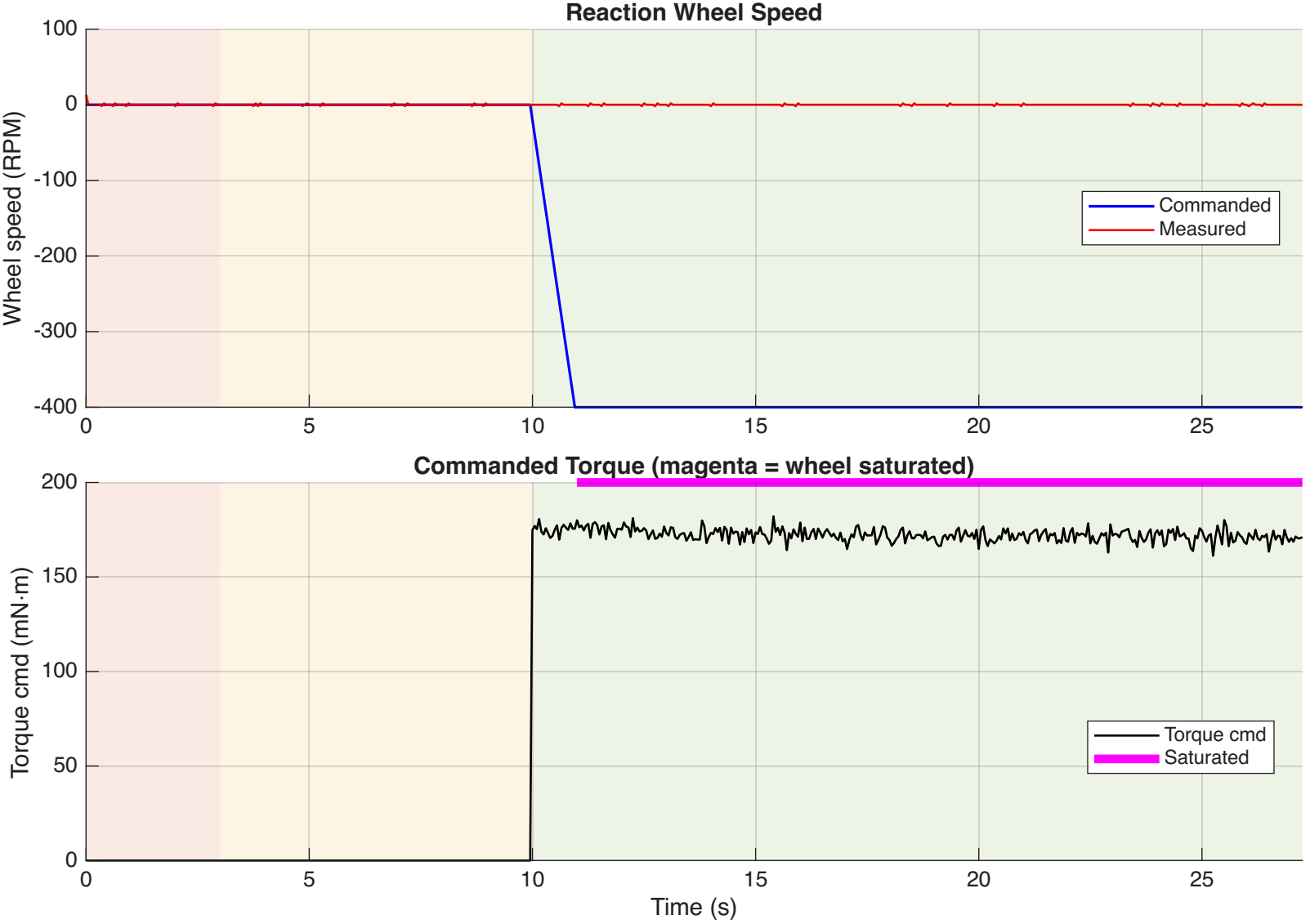
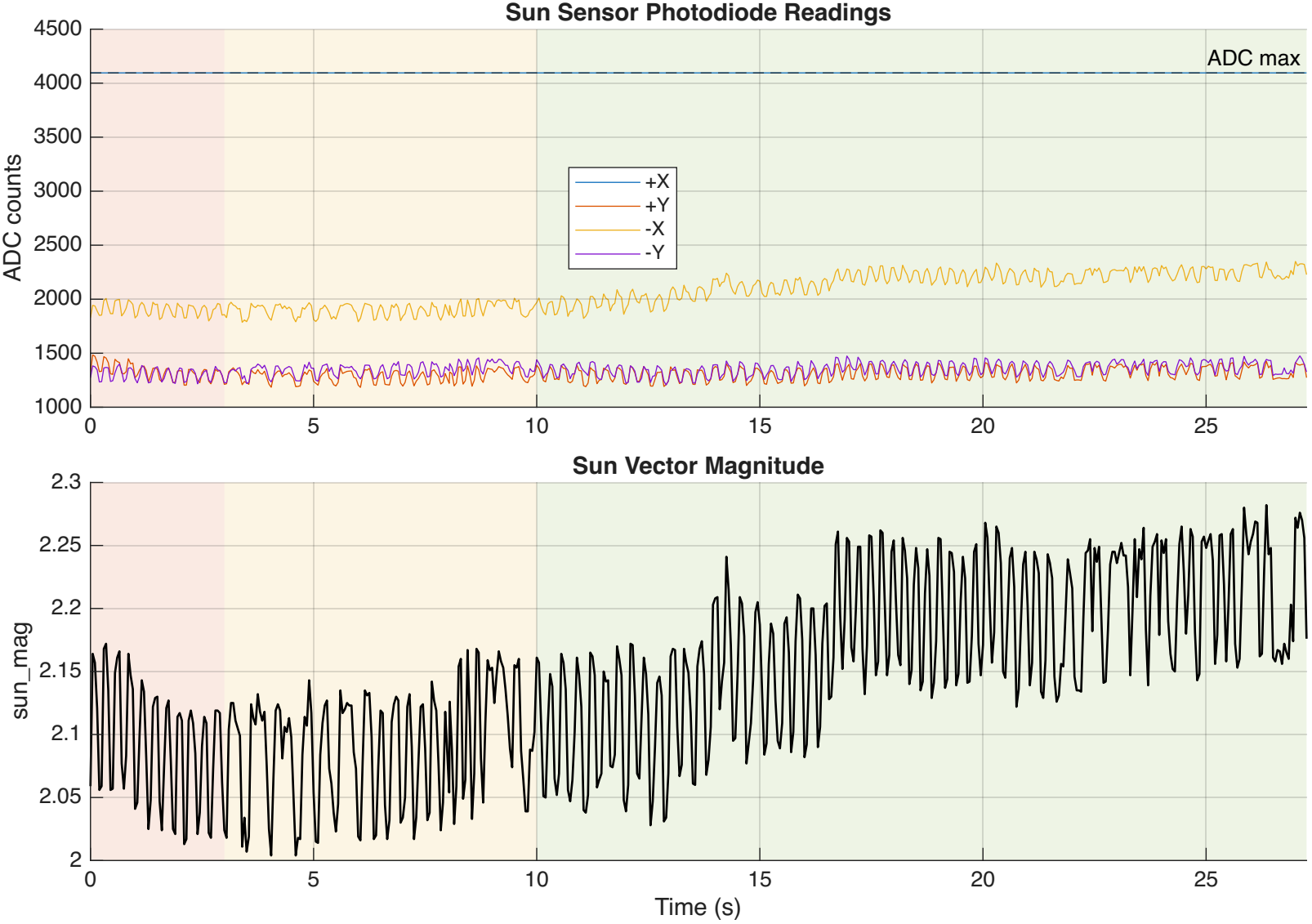
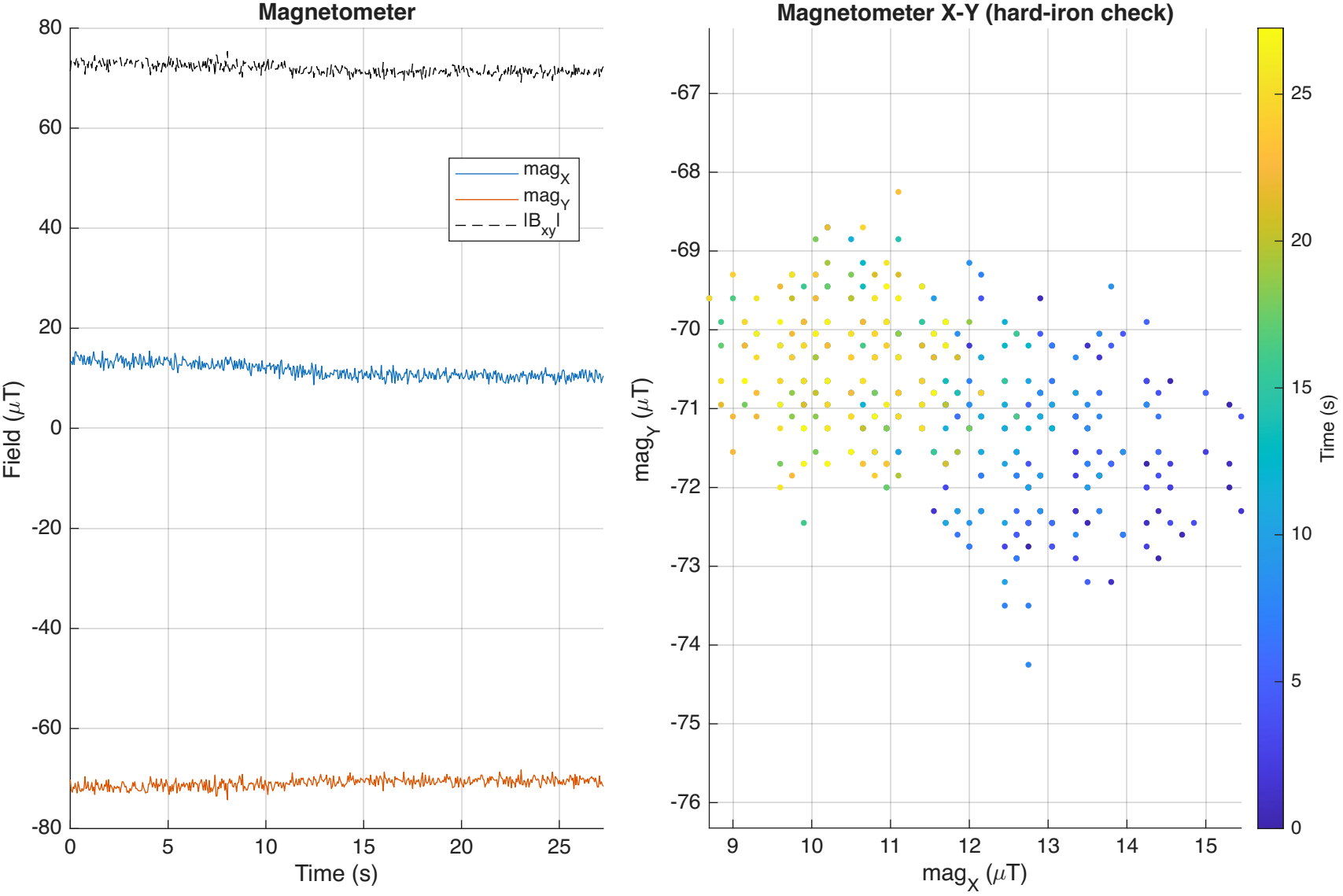
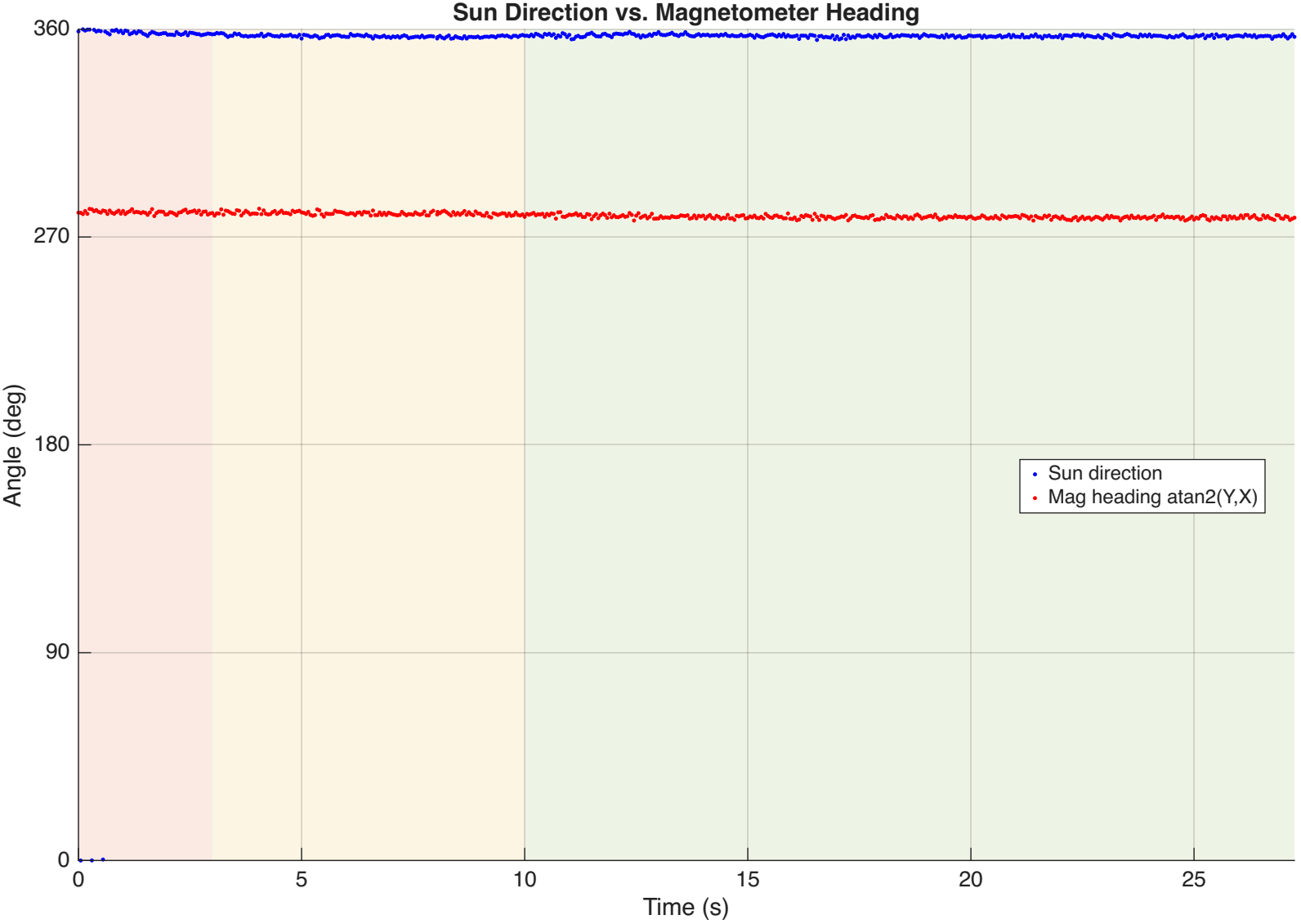

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
5
10
15
20
25
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
0
5
10
15
20
25
Time (s)

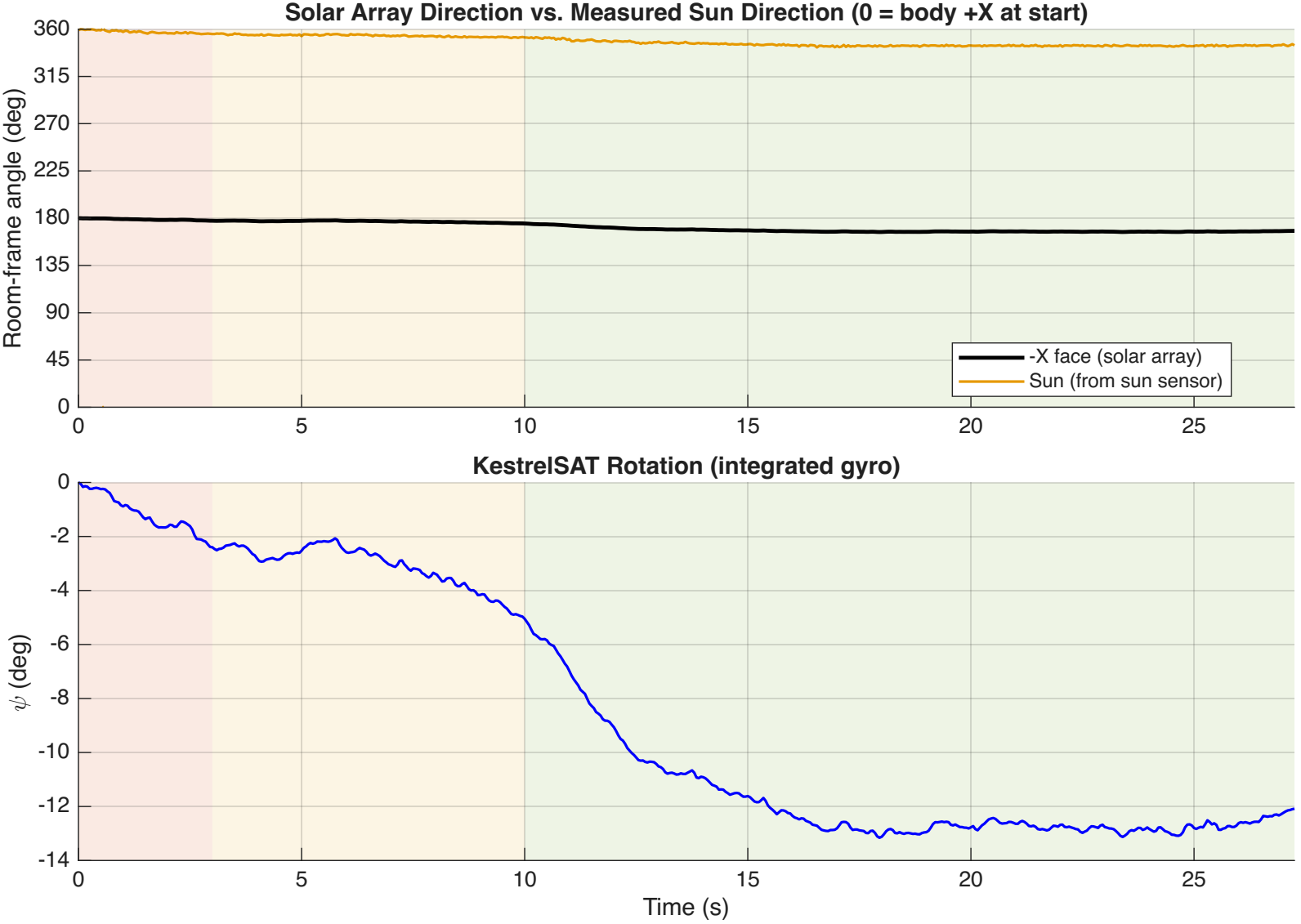

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
Time (s)
6
4
2
0
-2
-4
-6
0
5
10
15
20
25

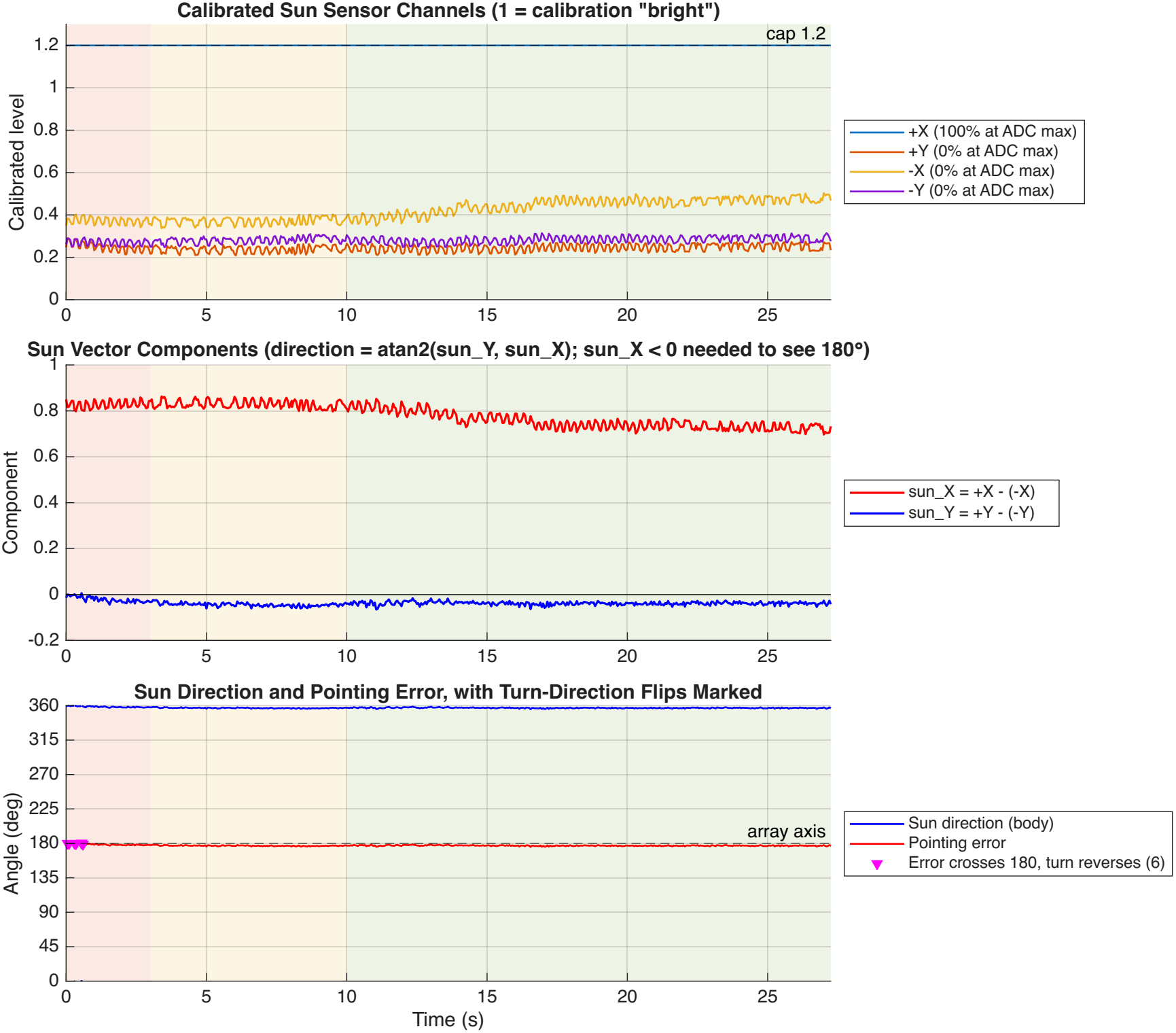

Reaction Wheel Speed
Wheel speed (RPM)
100
0
-100
-200
-300
-400
0
5
10
15
20
25
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
200
150
100
50
0
Torque cmd
Saturated
Time (s)

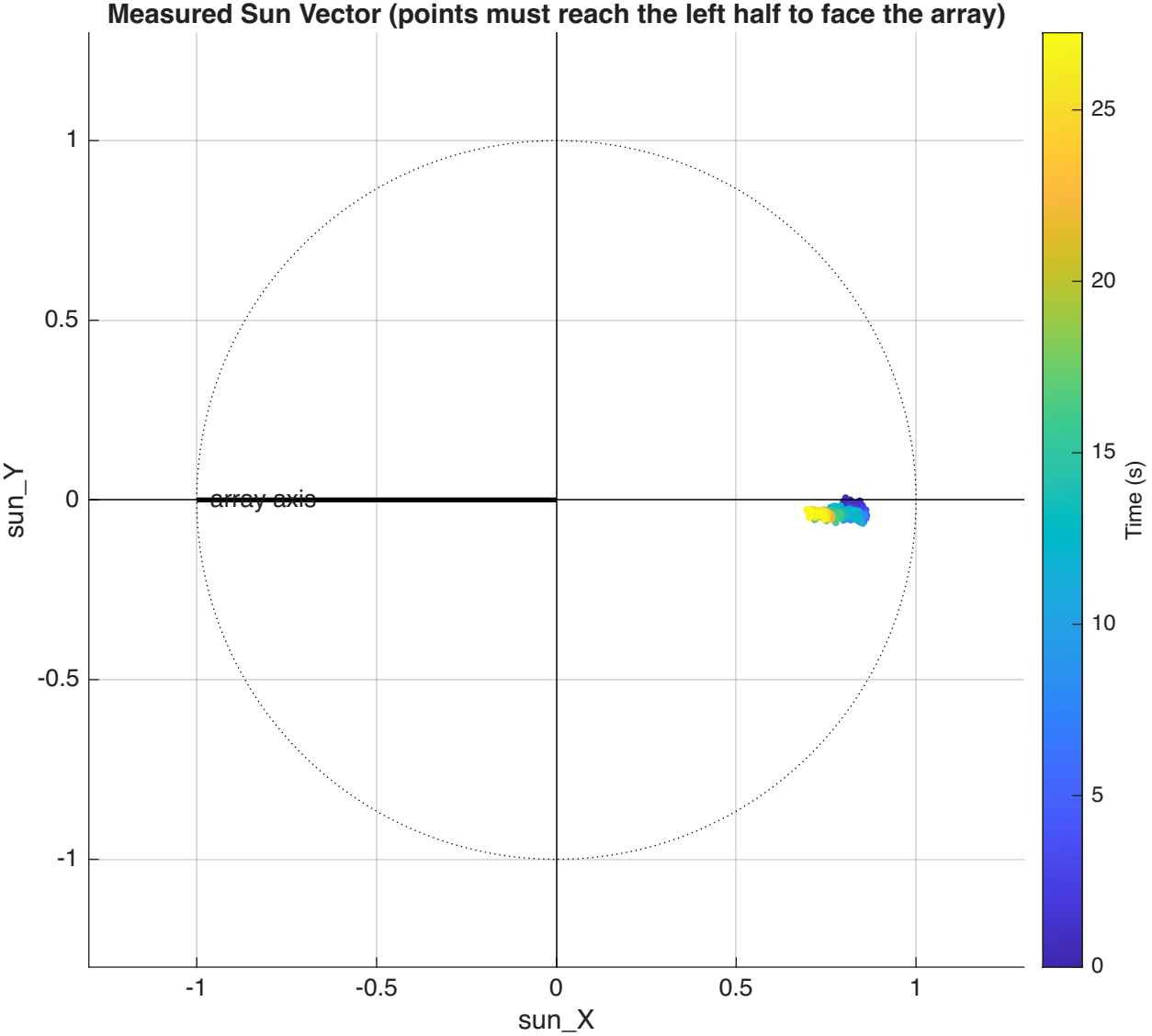

Sun Sensor Photodiode Readings
ADC max
+X
+Y
-X
-Y
ADC counts
4500
4000
3500
3000
2500
2000
1500
1000
0
5
10
15
20
25
Sun Vector Magnitude
sun_mag
2.3
2.25
2.2
2.15
2.1
2.05
2
Time (s)

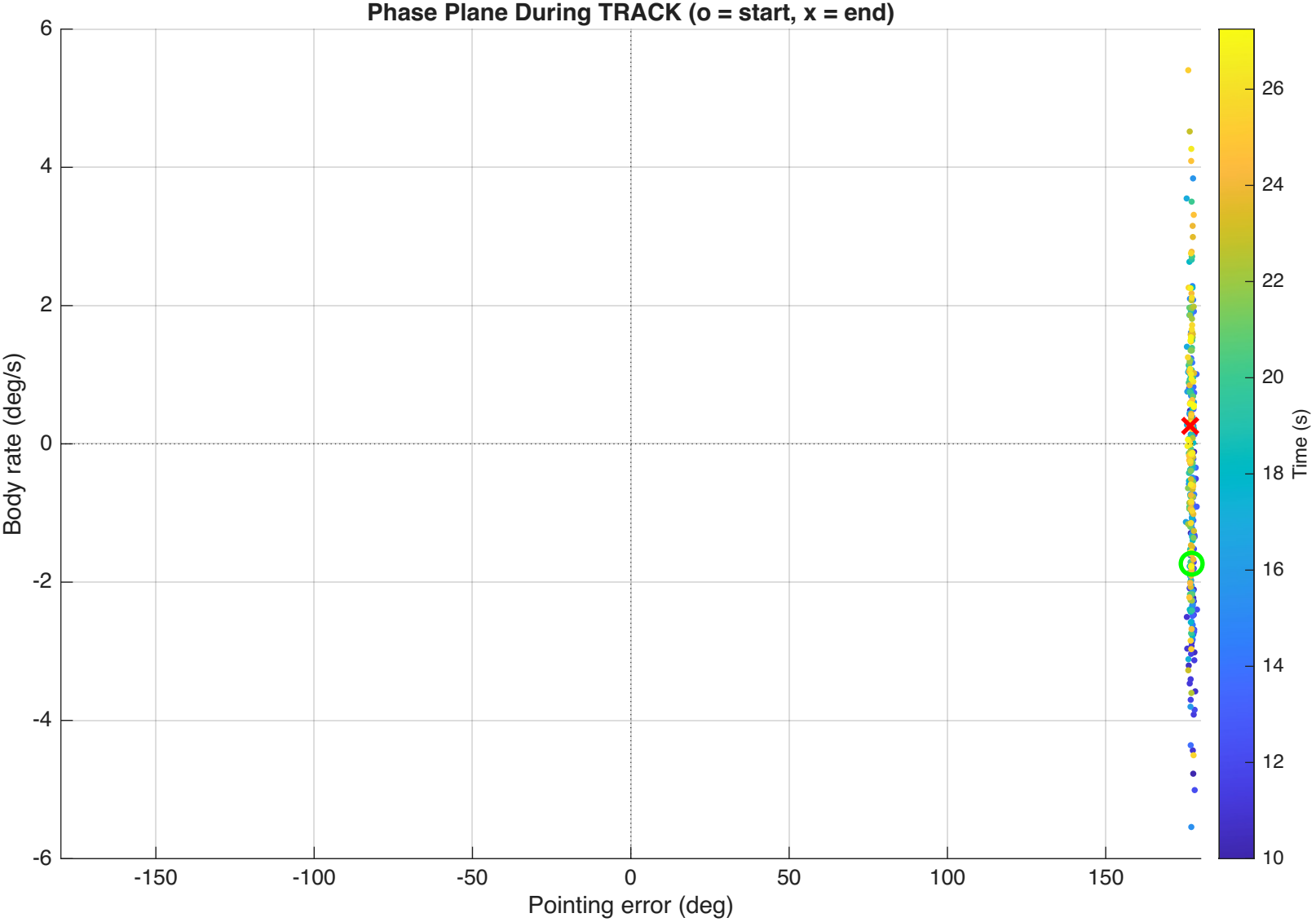

Magnetometer
Field (μT)
Time (s)
mag_X
mag_Y
|B_xy|
80
60
40
20
0
-20
-40
-60
-80
0
5
10
15
20
25
Magnetometer X-Y (hard-iron check)
mag_Y (μT)
mag_X (μT)
Time (s)
-67
-68
-69
-70
-71
-72
-73
-74
-75
-76
9
10
11
12
13
14
15
25
20
15
10
5
0

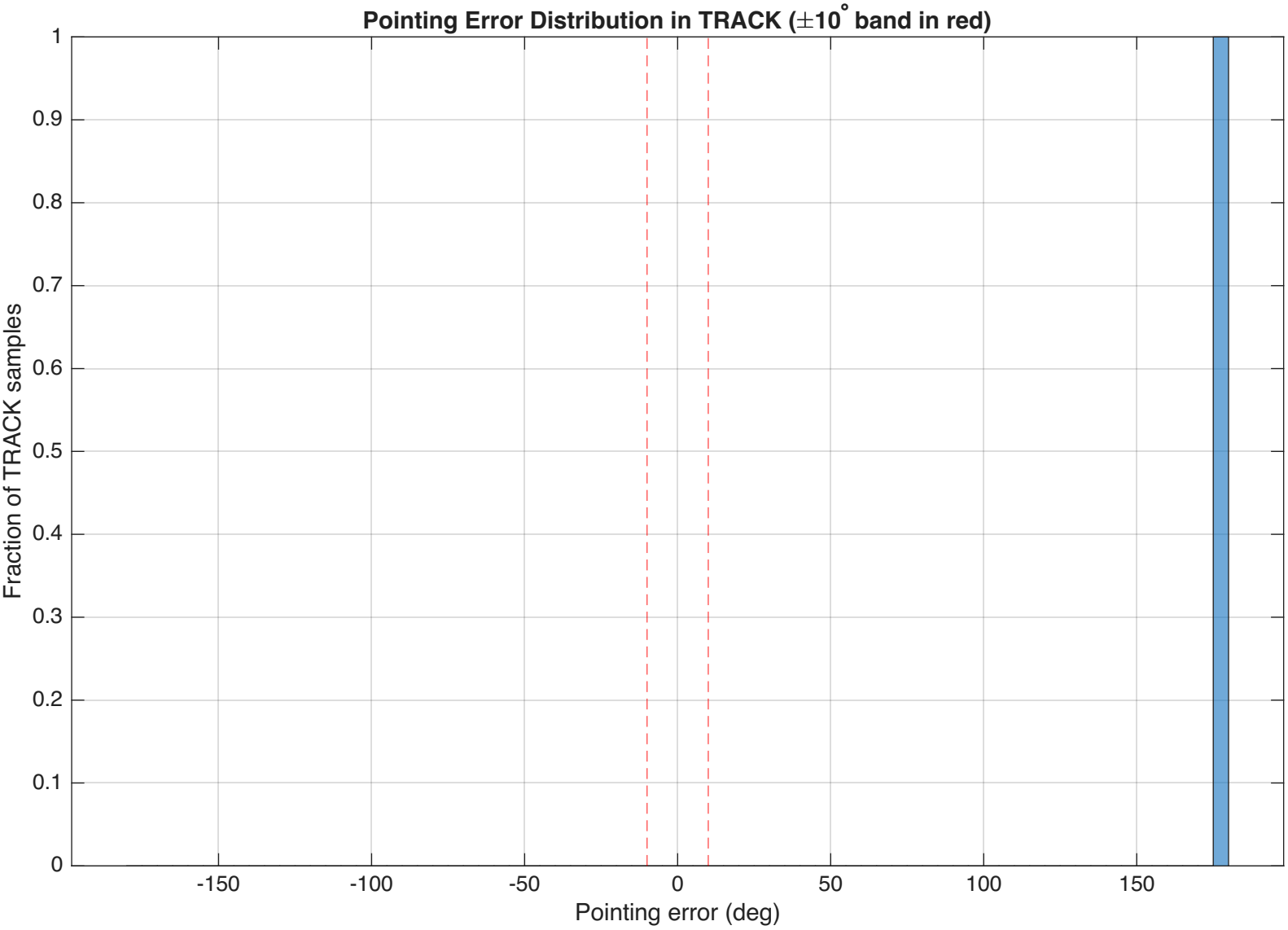

Sun Direction vs. Magnetometer Heading
360
270
180
90
0
Angle (deg)
0
5
10
15
20
25
Time (s)
Sun direction
Mag heading atan2(Y,X)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
5
10
15
20
25
-X face (solar array)
Sun (from sun sensor)
KestrelSAT Rotation (integrated gyro)
ψ (deg)
0
-2
-4
-6
-8
-10
-12
-14
0
5
10
15
20
25
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (100% at ADC max)
+Y (0% at ADC max)
-X (0% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Error crosses 180, turn reverses (6)
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
25
20
15
10
5
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
6
4
2
0
-2
-4
-6
-150
-100
-50
0
50
100
150
10
12
14
16
18
20
22
24
26


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-150
-100
-50
0
50
100
150